

Watershed Management to Strengthen Resilience to rain-fed Dry Spells in Gadaref State – Sudan

FAO's Technical Cooperation Programme (TCP)- TCP/SUD/3702

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Project partners

FAO	ICARADA	Gov. of Sudan ARC
■ Financial support ■ Technical backup and assistance	Technical support	• Local component • Field implementation and monitoring

**Project
Contribution to
Government and
FAO priorities:**

- **Government priority 1:** Food security
- Government priority 2: Sustainable Agriculture (Crops, Livestock, Fisheries, Forestry) and Natural Resources Management
- Government priority 3: Disaster Risk Management and Resilience Building
- **Regional initiatives:**
 - Regional Initiative on Water Scarcity for the Near East and North Africa (WSI).
 - Building Resilience for Food Security and Nutrition in the Near East and North Africa.
 - Small-Scale Family Farming Initiative for Inclusive Development in the Near East and North Africa.

Rationale and constraints

Rationale

- Gadaref State is predominately-agricultural state depending largely on rain-fed agriculture.
- Major food production and crops export area in the country
- Annual rainfall (300 – 600 mm) is often above the crop water requirement (400 -500 mm).
- Many crops fail or produce lower yield than the potential due to existing land pattern practices characterized by light tillage with mono-cropping and low inputs.

Constraints

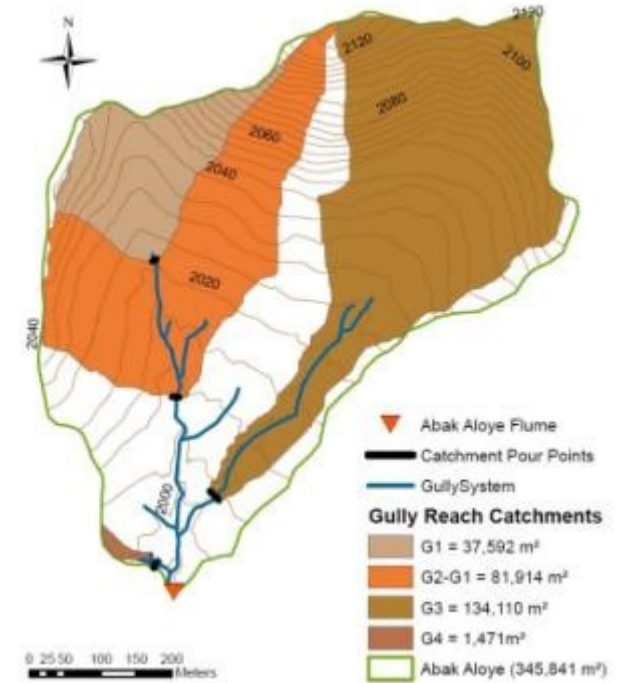
- low soil fertility, soil erosion, drought,
- Weed infestation due to mono-cropping,
- Heavy cracking soils, water logging and late planting
- Lack of soil and water conservation practices
- Poor access to advisory and extension services.

Impact of constraints

- 
1. Low crop yields
 2. Undermining the state's agricultural sector potential to address the government priority to achieve food security
 3. Increased vulnerability to climate change, water scarcity and land degradation and other factors

Aim of applied Research

Introduce and test integrated WSM in order to conserve and maximize resources use to increase productivity, reduce soil loss and improve ecosystem set up.



Project location and land use



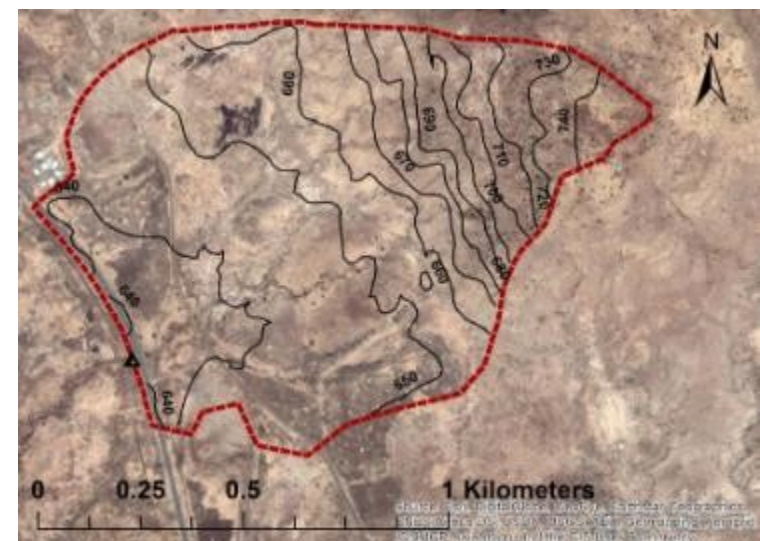
Traditional rainfed farming with apparent drainage lines & undulated surface



Watershed biophysical baseline & monitoring

Topography & Land use

- Field inspection and hand-held GPS survey
- Socio-economic survey and farmer's feedback on agricultural and cropping pattern
- Remote Sensing online sources (Aster v2 for DEM generation) for analysis of land topography
- Google Earth Pro images for land use and implementation-area definition



Assessment of land use and topography & GIS mapping

Watershed biophysical baseline & monitoring

Soil (1)

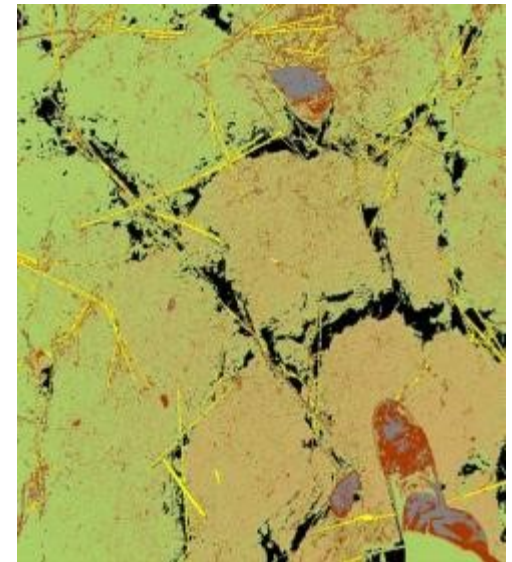
- Detailed soil sampling campaign designed by ICARDA
- Soil sampling performed by ARC
 - Soil physical properties (texture, bulk density, ...)
 - Soil chemical properties (pH, nutrients, ...)
- Monitoring of soil-crack development over rainy season 2016
 - Surface photography (1m^2) and exploring surface structure of soil cracks by means of GIS analysis.
 - Manual assessment using steel mesh
 - Manual measurement using a steel pin for depths
- Monitoring of gully development (width and areal extent)



Soil-cracks accelerating gully development



Soil profile sampling



Soil-crack image analysis

Cracks measurement



Project overall goal and objectives

Enhancing agricultural water productivity to contribute to food security and improve resilience and livelihood through adaptive watershed management practices:

Activities:

- i) Treatment of the landscapes to reduce deep cracks extend responsible for significant deep-water losses,
- ii) Control of surface runoff water for direct in-situ soil and water conservation use (contour plowing, planting pits, conservation agriculture, and afforestation);
- iii) Indirect collection and storage of ex-situ rainwater harvesting in farm ponds earth reservoirs,
- iv) Re-cycling this extra water for supplementary irrigation to bridge the inter-season rain dry spells effects.

Watershed activities implementation

Actual stage implementation

AWM

- Different field preparation techniques (contour mapping, earth embankments, contour plowing)
- Monitoring soil-cracking, crop yield, and soil water balance
- Construction of earth reservoirs (500-1000m³);
- digging 2 shallow wells
- Construction of small water collecting pits (ponds) for forest rehabiliattion and apiculture development



Construction of farm-ponds for supplemental irrigation

Watershed activities implementation



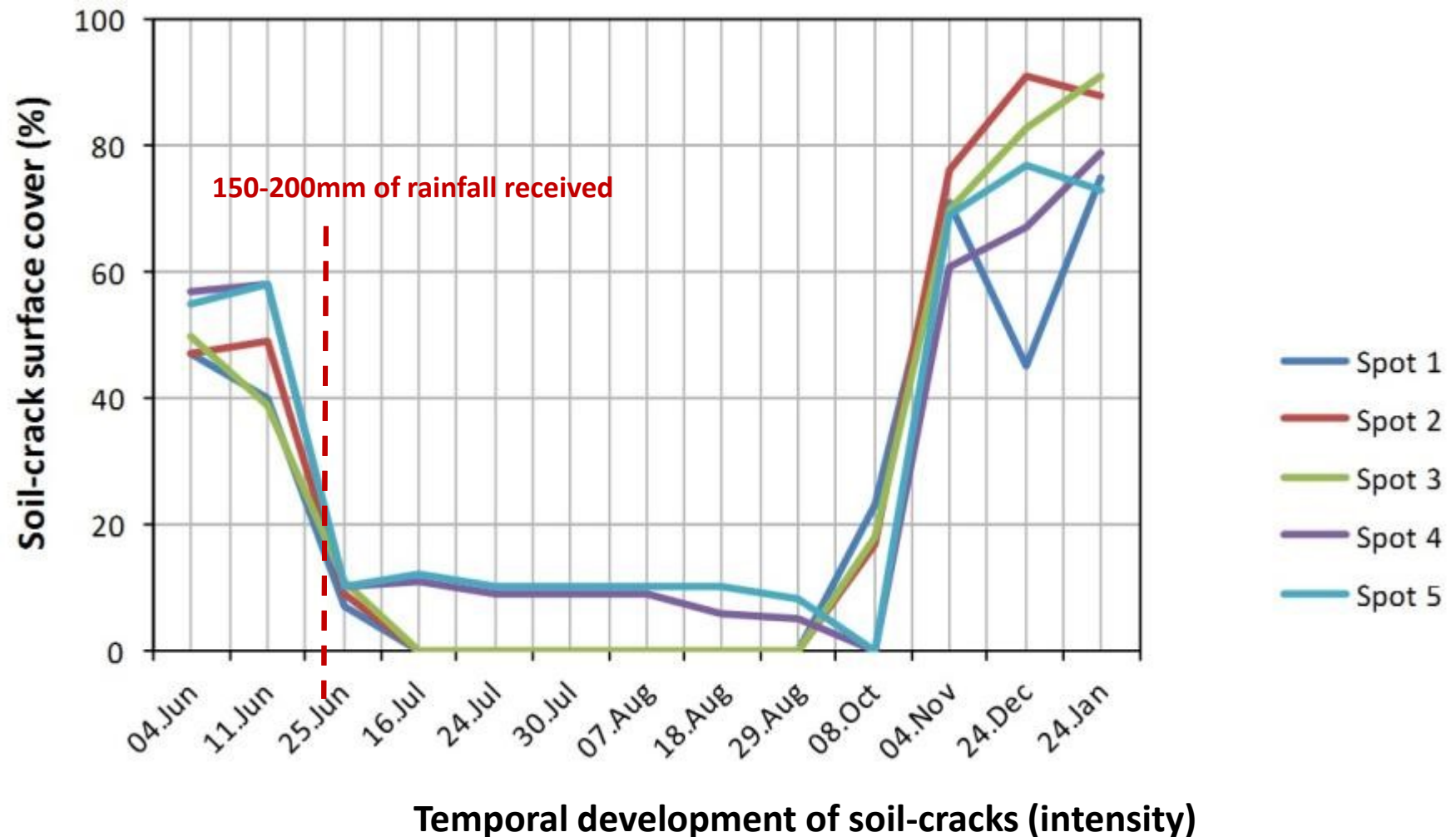
Results of the Watershed Management Trials



Watershed biophysical baseline & monitoring

Soil (2)

Spot 1,2 & 3: treated land – Spot 4 & 5: non-treated



Clay farm pond for supplementary irrigation and bee-keeping – Watershed in Gadaref

storage of ex-situ rainwater
harvesting



Rehabilitation of degraded forest WS with wet ponds for bee-keeping- Gadaref



Watershed implementation activities



5 months old acacia tree seedlings raised for Apiculture

Vegetative degraded area with cidir trees



Supplementary irrigation



Integrated crop management



Scaling up on farmers' fields

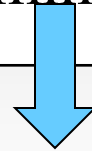


This TCP scaling
up demo plots
focus:

- Improved rain-fed crop productivity,
- Natural resources management and sustainable use (water, land and plant),
- Support small-scale rain-fed farmers through integrated soil and water conservation and cropping patterns
- Diversification of household income generating activities with women vegetable production and apiculture activities.
- Specifically contributes to the achievement of the Sustainable Development Goals relating to poverty, food and water security, climate change adaptation, and land conservation (SDGs 1, 2, 6, 13 and 15).

Expected Results

IMPACT	Enhanced food security, income generation, resilience of rural households to CC and environmental hazards in Gadaref State.
OUTCOME	Enhanced rain-fed farming water productivity



Indicator: Improved in crop productivity of rain-fed agriculture



Expected Results

OUTPUT 1	Agricultural watershed management (AWM) improvement technologies and innovations adopted
INDICATOR	Landscapes management in farmers' fields established to show the advantages of the technologies being promoted compared to the farmers practices

ACTIVITIES:

- Conduct inception workshop
- Introduction of project concept to stakeholders,
- Reconnaissance survey for preliminary sites investigation & selection,
- Farmers animation and selection
- Design and installation of WS management technologies and monitoring



Expected Results

OUTPUT 2	Capacity of stakeholders for adopting sustainable AWM Improvement technologies strengthened and enhanced
INDICATOR	The skills of producers and implementing partners is capacitated to use and promote gender sensitive appropriate AW Management technologies and practices

ACTIVITIES:

- ToT training (extension staff and water engineers & technicians), FFS training
- Joint “extension-farmers” tours to demo sites and field days
- Regional study tour
- Regional workshops /meetings





Farmers and community training

Community	Men	Women	Total
Abu hamra	43	53	96
Wad Elshagra	126	141	267
Greeb and Doka	80	320	400
Total	249	514	763

Expected Results

OUTPUT 3	Knowledge on sustainable AWM and food security improvement disseminated
INDICATOR	Knowledge base on AWM produced and disseminated by effective learning alliances (leaflets, pamphlets, manuals, web-site)

ACTIVITIES:

- ✓ Improving existing learning network at ARC for dissemination and promotion of project knowledge products
- ✓ Production of project-based results knowledge products (KPs)
- ✓ Final Workshop

Sunflower under traditional farming



Sunflower under well developed land
(contour plowing ,earth
embankments & chiseling)



GN with well developed rain-fed land
& improved crop husbandry
management



GN with traditional practices



Mechanized land treatment is suited to relatively large areas



Farmers average crop yields under different SWC practices (2020)

No. famers	Crop	Yield (t/ha)			
		Chiseling	Chiseling+ Contour Plowing	WLD*	WLD+ earth dikes
10	sorghum	1.13	1.82	0.42	0.8
7	groundnut	1.6	1.8	1.0	-
15	Sunflower	0.6	0.90	0.5	-

* WLD: wide level Disc



Recommendations

- **Increasing loans ceiling to small producer's rain-fed farmers**
- **Increasing awareness about investment and rewards on land development of undulating surface and clay soil**
- **Continuous capacity building**



Thanks
I Wish you Great Day

